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Tool for a connector assembly

Abstract

A tool for a connector assembly includes a crimp head having a tool housing and a ram assembly disposed within the tool housing, the ram assembly has a connector ram and a shield ram disposed within the connector ram. The tool includes a driving portion driving movement of the ram assembly with respect to the tool housing from a retracted position to a terminated position along a longitudinal direction. The connector ram and the shield ram are movable together with respect to the tool housing from the retracted position to an intermediate position between the retracted position and the terminated position. The shield ram is movable with respect to the connector ram from the intermediate position to the terminated position of the ram assembly.

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Background/Summary

FIELD OF THE INVENTION

(1) The present invention relates to a terminating assembly including a tool used for terminating a connector assembly.

BACKGROUND

(2) A wire is often electrically connected to a printed circuit board (PCB) by a connector disposed on the PCB. The connector is commonly soldered to the PCB. The wire is inserted into the connector and, in insulation displacement contact applications, the connector is terminated by pressing a cover of the connector toward the PCB. The cover presses the wire against a contact in the connector to electrically connect the wire with the contact and, through the solder joint, with the PCB.

(3) The insertion force required to displace the cover of the connector and terminate the wire is commonly dispersed to the PCB. The force reaching the PCB, however, can lead to damage of the solder joint between the connector and the PCB, impairing the electrical connection to the PCB. Further, terminating the wire in the connector by pressing the cover either manually or with existing tools is inefficient and increases production costs.

SUMMARY

(4) A tool for a connector assembly includes a crimp head having a tool housing and a ram assembly disposed within the tool housing, the ram assembly has a connector ram and a shield ram disposed within the connector ram. The tool includes a driving portion driving movement of the ram assembly with respect to the tool housing from a retracted position to a terminated position along a longitudinal direction. The connector ram and the shield ram are movable together with respect to the tool housing from the retracted position to an intermediate position between the retracted position and the terminated position. The shield ram is movable with respect to the connector ram from the intermediate position to the terminated position of the ram assembly.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The invention will now be described by way of example with reference to the accompanying Figures, of which:
- (2) FIG. 1 is a perspective view of a terminating assembly according to an embodiment;
- (3) FIG. 2 is an exploded perspective view of a tool of the terminating assembly;
- (4) FIG. 3 is a sectional perspective view of a crimp head of the tool;
- (5) FIG. 4 is a side view of a driving portion of the tool with a portion of a handle removed;
- (6) FIG. 5 is a perspective view of a connector assembly of the terminating assembly;
- (7) FIG. 6 is a sectional perspective view of the connector assembly;
- (8) FIG. 7A is a plan view of the terminating assembly with a clamp assembly of the tool in an open position;
- (9) FIG. 7B is a plan view of the terminating assembly with the clamp assembly in a closed position;
- (10) FIG. 8 is a sectional perspective view of the terminating assembly with a ram assembly of the tool in an intermediate position;
- (11) FIG. 9 is a sectional perspective view of the terminating assembly with the ram assembly in a terminated position;
- (12) FIG. 10A is a side view of the driving portion with a ratchet assembly and a portion of a handle removed;
- (13) FIG. 10B is a side view of the driving portion with the ratchet assembly in the terminated position; and
- (14) FIG. 11 is a perspective view of the tool and the connector assembly after termination of the connector assembly.

DETAILED DESCRIPTION OF THE EMBODIMENTS

(15) Exemplary embodiments of the present disclosure will be described hereinafter in detail with reference to the attached drawings, wherein like reference numerals refer to like elements. The present disclosure may, however, be embodied in many different forms and should not be construed as being limited to the embodiments set forth herein; rather, these embodiments are provided so that the present disclosure will convey the concept of the disclosure to those skilled in the art. In addition, in the following detailed description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed embodiments. However, it is apparent that one or more embodiments may also be implemented without these specific details.

(16) Throughout the specification, directional descriptors are used such as “longitudinal” and “width”. These descriptors are merely for clarity of the description and for differentiation of the various directions. These directional descriptors do not imply or require any particular orientation of the disclosed elements.

(17) Throughout the drawings, only one of a plurality of identical elements may be labeled in a figure for clarity of the drawings, but the detailed description of the element herein applies equally to each of the identically appearing elements in the figure.

(18) A terminating assembly **10** according to an embodiment, as shown in FIG. 1, includes a tool **100** and a connector assembly **200**. The tool **100**, as described in detail below, is used to terminate the connector assembly **200**.

(19) As shown in FIG. 1, the tool **100** includes a crimp head **110** and a driving portion **180** attached to the crimp head **110** and driving movement of the crimp head **110**. The crimp head **110**, as shown in FIGS. 1-3, includes a tool housing **120**, a ram assembly **130** disposed within the tool housing **120**, and a clamp assembly **160** attached to the tool housing **120**.

(20) The tool housing **120**, as shown in FIGS. 2 and 3, includes a tool body **122** extending from a first body end **123** to a second body end **124** along a longitudinal direction L. The tool body **122** has a housing passageway **125** extending through the tool body **122** from the first body end **123** to the second body end **124**.

(21) As shown in FIGS. 2 and 3, the tool housing **120** has a pair of side plates **126** attached to an outer side of the tool body **122** adjacent to the second body end **124**. Each of the side plates **126** has a slot **127** extending through the side plate **126** in a width direction W perpendicular to the longitudinal direction L. The slot **127** is elongated in the side plate **126** along the longitudinal direction L. Each of the side plates **126** has a hinge **128** at an end of the side plate **126** adjacent to the second body end **124**.

(22) The ram assembly **130**, as shown in FIGS. 2 and 3, includes a connector ram **132** and a shield ram **134** disposed within the connector ram **132**. The connector ram **132** extends from a spring end **133** to a cap end **134** along the longitudinal direction L and has a ram passageway **135** extending through the connector ram **132**, as shown in FIG. 3. The ram passageway **135** bends along the longitudinal direction L. The shield ram **140** is disposed within the ram passageway **135** and, between a rear end **142** and a shield end **144** opposite the rear end **142** along the longitudinal direction L, the shield ram **140** has a shape corresponding to the shape of the ram passageway **135**, as shown in FIG. 3.

(23) As shown in FIGS. 2 and 3, the ram assembly **130** includes an inserter rod **146** attached to the rear end **142** of the shield ram **140** and extending away from the shield ram **140** along the longitudinal direction L. The ram assembly **130** includes a shield ram adjustor **148** attached to an end of the inserter rod **146** opposite the shield ram **140** and a preload adjustor **150** disposed around the inserter rod **146**. The shield ram adjustor **148** is a cylindrical element having a flange **149**. The shield ram adjustor **148** is attachable to the end of the inserter rod **146** in a plurality of different positions along the longitudinal direction L; in an embodiment, the shield ram adjustor **148** is attached by a threaded connection to the inserter rod **146** to allow for adjustment of a distance of the flange **149** from the inserter rod **146** along the longitudinal direction L. The preload adjustor **150** is an annular element disposed around the inserter rod **146** and is movable to a plurality of different positions along the inserter rod **146**; in an embodiment, the preload adjustor **150** is attached by a threaded connection to the inserter rod **146** to allow for adjustment of a position of the preload adjustor **150** along the longitudinal direction L of the inserter rod **146**.

(24) As shown in FIG. 3, the ram assembly **130** is positioned in the housing passageway **125** of the tool body **122**. The connector ram **132** and the shield ram **140** disposed within the connector ram **132** are positioned within the housing passageway **125** adjacent to the second body end **124**. The inserter rod **146** extends along the housing passageway **125** and the shield ram adjustor **148** connected to the inserter rod **146** is positioned adjacent to the first body end **123** of the tool body **122**. The preload adjustor **150** is disposed within the housing passageway **125** and, as shown in FIG. 3, has an outer dimension approximately equal to a diameter of the housing passageway **125**.

(25) The ram assembly **130**, as shown in FIGS. 2 and 3, includes a return spring **152** disposed between the first body end **123** and the flange **149** of the ram adjustor **148** and a preload spring **154** disposed in the housing passageway **125** between the preload adjustor **150** and the spring end **133** of the connector ram **132**.

(26) As shown in FIGS. 2 and 3, the ram assembly **130** includes a plurality of ram stops **156** disposed within the tool housing **120** and extending into the housing passageway **125**. The ram stops **156** are each a cylindrical member having a pointed end **157** positioned in the housing passageway **125**. In an embodiment, a position of the ram stops **156** with respect to the tool housing **120**, and a corresponding protruding distance of the pointed end **157** into the housing passageway **125**, is adjustable. In an embodiment, the ram stops **156** are threaded and positioned in a threaded opening of the tool housing **120** to permit adjustment of the positioning of the ram stops **156**.

(27) The ram assembly **130** includes a pair of jaw plates **158**, shown in FIG. 2. The jaw plates **158** are each an approximately rectangular member extending in the longitudinal direction L and having a leading end **159**. The jaw plates **158** are attached to the connector ram **132** by a plurality of fasteners **102** that extend through the slots **127** of the side plates **126**. In the shown embodiment, the fasteners **102** are bolts that attach the jaw plates **158** to the connector ram **132**. In other embodiments, the fasteners **102** can be any other type of fastening element that can attach the jaw plates **158** to the connector ram **132** and extend through the slots **127** of the side plates **126**.

(28) The clamp assembly **160**, as shown in FIGS. 1-3, includes a pair of jaws **162** pivotably attached to the tool housing **120**. Each of the jaws **162** extends from a pivot end **163** to a hook **164** along the longitudinal direction L. The hook **164** of each of the jaws **162** has an L-shape and bends to extend in the width direction W perpendicular to the longitudinal direction L.

(29) As shown in FIGS. 2 and 3, the clamp assembly **160** includes a pair of pins **170** each attaching one of the jaws **162** to one of the side plates **126** of the tool housing **120**. The pins **170** each extend through the pivot end **163** of one of the jaws **162** and the hinge **128** of one of the side plates **126**. The jaws **162** are each pivotable about the pin **170** and the hinge **128** of the side plate **126**.

(30) The clamp assembly **160**, as shown in FIGS. 2 and 3, includes a plurality of biasing springs **172**. As shown in FIG. 3, the biasing springs **172** each have a first end **174** and a second end **176** opposite the first end **174** along the longitudinal direction L. The first end **174** of each of the springs **172** is disposed in the second body end **124** of the tool body **122**. The second end **176** of each of the springs **172** is disposed in the pivot end **163** of one of the jaws **162**.

(31) The driving portion **180**, in the embodiment shown in FIGS. 1 and 2, has a handle **182** and a trigger **186** disposed within the handle **182** that is movable within the handle **182**. As shown in FIG. 4, the handle **182** has a handle passageway **184** in which the trigger **186** is partially disposed. The trigger **186** is pivotable about a trigger axis **188** positioned in the handle passageway **184**.

(32) Movement of the trigger **186**, for example by manual actuation from a user, pivots the trigger **186** about the trigger axis **188**; movement of the trigger **186** with respect to the handle **182** drives movement of the ram assembly **130** of the crimp head **110**, as described in greater detail below. The handle **182** and the trigger **186** manually actuated by a user, however, are only one embodiment of the driving portion **180**. In other embodiments, the driving portion **180** may be an arbor press, a pneumatic hand tool, or any other element capable of actuating the ram assembly **130** as described herein.

(33) In the embodiment shown in FIG. 4, the driving portion **180** has a ratchet assembly **190** that restricts pivoting of the trigger **186** about the trigger axis **188** in certain positions. The ratchet assembly **190** includes a ratchet tab **191** extending from the trigger **186**. On a side distal from the trigger **186**, the ratchet tab **191** has a first sloped side **192**, a central surface **193** extending from the first sloped side **192**, and a second sloped side **195** extending from a side of the central surface **193** opposite the first sloped side **192**. The central surface **193** has a plurality of teeth **194**. A trigger spring **187** is connected to the ratchet tab **191** and biases the trigger **186** to a retracted position R about the trigger axis **188** shown in FIG. 4.

(34) As shown in FIG. 4, the ratchet assembly **190** includes a pawl **196** attached to the handle **182** in the handle passageway **184**. The pawl **196** is pivotable about a pawl axis **198** and has a point **197** at an end of the pawl **196**. A pawl spring **199** connected to the pawl **196** biases the pawl **196** into a position about the pawl axis **198** in which the point **197** is adjacent to the ratchet tab **191**.

(35) The connector assembly **200** is shown in detail in FIGS. 5 and 6. The connector assembly **200** includes a printed circuit board **210** (hereinafter, "PCB"), a connector **220** disposed on the PCB **210**, and a wire **260** connected to the PCB **210** by the connector **220**.

(36) The connector **220**, as shown in FIGS. 5 and 6, has a connector housing **222**, a contact **230** disposed in the connector housing **222**, a cap **240** disposed on the connector housing **222**, and a shield **250** disposed on the connector housing **222**.

(37) As shown in FIG. 5, the connector housing **222** is attached to an upper surface **212** of the PCB

210 and, as shown in FIG. 6, defines a receiving space **226** within the connector housing **222**. The connector housing **222** has a pair of projections **224** extending from the connector housing **222** in the width direction W. The projections **224** extend approximately parallel to the upper surface **212** of the PCB **210** and are spaced apart from the upper surface **212** in the longitudinal direction L, as shown in FIG. 5. In an embodiment, the connector housing **222** is monolithically formed in a single piece of insulative material with the projections **224**.

(38) The contact **230**, as shown in FIG. 6, is disposed in the connector housing **222** and protrudes into the receiving space **226**. The contact **230** is formed of a conductive material and, in the shown embodiment, is an insulation displacement contact.

(39) The cap **240** is shown in a cap protruding position CP on the connector housing **222** in FIGS. 5 and 6. In the cap protruding position CP, the cap **240** is spaced apart from the upper surface **212** of the PCB **210** in the longitudinal direction L. The wire **260** is disposed within the receiving space **226** of the connector housing **222** under the cap **240** and, in the cap protruding position CP, the wire **260** is spaced apart from the contact **230** disposed in the connector housing **222**. The wire **260** includes an insulation **262** disposed around a conductor **264** and a shield layer **266** disposed between the insulation **262** and the conductor **264**.

(40) The shield **250** is shown in a shield protruding position SP on the connector housing **222** in FIGS. 5 and 6. In the shield protruding position SP, the shield **250** is spaced apart from a portion of the cap **240** and spaced apart from the wire **260** along the longitudinal direction L.

(41) A process of using the tool **100** to terminate the connector assembly **200** will now be described in greater detail primarily with reference to FIGS. 7A-11. The connector assembly **200** begins with the cap **240** in the cap protruding position CP and the shield **250** in the shield protruding position SP, as shown in FIGS. 5 and 6.

(42) The tool **100** starts in the retracted position R shown in FIGS. 1 and 7A. In the retracted position R, the jaws **162** of the clamp assembly **160** are in an open position O. As shown in FIG. 3, the biasing springs **172** bias the jaws **162** toward the open position O, providing a biasing force that pivots the jaws **162** about the pins **170** away from the second body end **124** of the tool body **122**. The ram assembly **130** is positioned in the tool housing **120** as shown in FIG. 3 in the retracted position R, with the leading ends **159** of the jaw plates **158**, as shown in FIG. 7A, aligned with the pins **170** and the hinges **128** of the side plates **126** along the longitudinal direction L.

(43) With the jaws **162** in the open position O and the driving portion **180** and the ram assembly **130** in the retracted position R, the tool **100** is moved toward the connector assembly **200** along the longitudinal direction L as shown in FIG. 7A. The tool **100** is moved along the longitudinal direction L toward the connector **220** and the PCB **210** until the hooks **164** of the jaws **162** pass beyond the projections **224** of the connector housing **222** in the longitudinal direction L.

(44) The driving portion **180** is then actuated to move the ram assembly **130** from the retracted position R shown in FIG. 7A to an intermediate position I shown in FIGS. 7B and 8. From the retracted position R to the intermediate position I, as shown for example by comparison of FIG. 3 to FIG. 8, the shield ram adjustor **148** moves along the longitudinal direction L under a force from the driving portion **180**. To move the ram assembly **130** out of the retracted position R, the force applied by the driving portion **180** must overcome a spring force of the return spring **152**, moving the shield ram adjustor **148** toward the first body end **123** of the tool body **122** along the longitudinal direction L. The return spring **152** biases the ram assembly **130** toward the retracted position R.

(45) Movement of the shield ram adjustor **148** along the longitudinal direction L correspondingly moves the inserter rod **146** along the longitudinal direction L. From the retracted position R shown in FIG. 3 to the intermediate position I shown in FIG. 8, the preload spring **154** remains in an extended state; movement of the inserter rod **146** along the longitudinal direction L correspondingly moves the connector ram **132** and the shield ram **140** along the longitudinal direction L in the tool body **122** to the intermediate position I.

(46) As the ram assembly **130** moves along the longitudinal direction L in the tool housing **120**, the jaw plates **158** connected to the connector ram **132** likewise move along the longitudinal direction L outside the side plates **126** of the tool housing **120**. The fasteners **102** that connect the jaw plates **158** to the connector ram **132** move within the slots **127** of the side plates **126** as the connector ram **132** moves along the longitudinal direction L. As shown in FIG. 7B, as the connector ram **132** moves toward the intermediate position I, the leading ends **159** of the jaw plates **158** abut the outer surfaces of the jaws **162** and move along the jaws **162** in the longitudinal direction L, pivoting the jaws **162** about the pins **170** against the bias of the biasing springs **172**.

(47) In the intermediate position I shown in FIG. 7B, the jaw plates **158** have moved along the jaws **162** in the longitudinal direction L to pivot the jaws **162** and the clamp assembly **160** from the open position O to a closed position C. In the closed position C, the jaws **162** engage the connector housing **222**, with the hook **164** of each of the jaws **162** engaging one of the projections **224** of the connector housing **222**. The hook **164** of each of the jaws **162** is disposed between one of the pair of projections **224** and the upper surface **212** of the PCB **210**; the jaws **162** bear on the connector housing **222** via the projections **224** in the closed position C.

(48) As shown in FIG. 8, with the jaws **162** engaged with the connector housing **222**, the connector ram **132** reaches the intermediate position I and presses the cap **240** toward the connector housing **222** from the cap protruding position CP to a cap seated position CS on the connector housing **222**. In the movement toward the intermediate position I, the cap end **134** of the connector ram **132** contacts the cap **140** and, as the connector ram **132** continues to move along the longitudinal direction L toward the intermediate position I, the connector ram **132** presses the cap **240** to the cap seated position CS. In moving to the cap seated position CS, the cap **240** presses the wire **260** into the contact **230**, terminating the wire **260** to the contact **230**; the contact **230** pierces through the insulation **262** of the wire **260** and electrically connects with the conductor **262** of the wire **260**. As shown in FIG. 11, the cap **140** abuts the upper surface **212** of the PCB **210** in the cap seated position CS.

(49) When the connector ram **132** reaches the intermediate position I shown in FIG. 8 in which the cap **140** is fully seated in the cap seated position CS, the connector ram **132** abuts the pointed ends **157** of the ram stops **156** that protrude into the housing passageway **125**. The ram stops **156** prevent the connector ram **132** from moving further along the longitudinal direction L toward the connector assembly **200** from the intermediate position I, preventing excessive force on the cap **140** and overcompression of the cap **140** on the connector housing **222**.

(50) Continued actuation of the driving portion **180** moves the ram assembly **130** from the intermediate position I shown in FIG. 8 to a terminated position T shown in FIG. 9. The jaws **162** remain engaged with the connector housing **122** as the ram assembly **130** moves from the intermediate position I to the terminated position T. Because the connector ram **132** abuts the ram stops **156**, continued force from the driving portion **180** that is greater than a preload force of the preload spring **154** moves the shield ram **140** within the ram passageway **135** of the connector ram **132** along the longitudinal direction L. The connector ram **132** remains in a same position in the tool housing **120** along the longitudinal direction L as the shield ram **140** moves from the intermediate position I to the terminated position T.

(51) As the shield ram **140** moves along the longitudinal direction L within the ram passageway **135** to the terminated position T shown in FIG. 9, the shield end **144** of the shield ram **140** contacts the shield **250** and presses the shield **250** to a shield seated position SS on the connector housing **222**. In the shield seated position SS, the shield **250** pierces the insulation **262** of the wire **260** and electrically contacts the shield layer **266** of the wire **260**. When the ram assembly **130** reaches the terminated position T, the wire **260** is fully terminated in the connector **220** and the connector **220** electrically connects the wire **260** to the PCB **210**.

(52) In the shown embodiment, the ratchet assembly **190** of the driving portion **180** restricts movement of the ram assembly **130** back to the retracted position R until the ram assembly **130**

reaches the terminated position T. As shown in FIG. 10A, once the ram assembly 130 moves out of the retracted position R under movement of the trigger 186, the point 197 of the pawl 196 engages the teeth 194 on the central surface 193 of the ratchet tab 191. In this position, if the driving force of the driving portion 180 is removed, such as by a user releasing pressure on the trigger 186, the pawl 196 engages the teeth 194 and prevents the trigger 186 from rotating back toward the retracted position R around the trigger axis 188.

(53) When the ram assembly 130 reaches the terminated position T under driving of the driving portion 180, as shown in FIG. 10B, the pawl 196 reaches the second sloped side 195 of the ratchet tab 191. The point 197 of the pawl 196 is no longer engages with the teeth 194 of the central surface 193 in the terminated position T and the trigger 186 is free to rotate back toward the retracted position R about the trigger axis 188.

(54) As shown in FIG. 11, once the ram assembly 130 reaches the terminated position T and the connector assembly 200 is fully terminated, the driving portion 180 releases the driving force and the ram assembly 130 moves back to the retracted position R. With the jaw plates 158 retracted, the biasing springs 172 bias the jaws 162 into the open position O and the tool 100 can be removed from engagement with the connector assembly 200.

(55) The clamp assembly 160 of the tool 100 according to the invention bears on the connector housing 222 while the tool 100 moves the cap 240 to the cap seated position CS and the shield 250 to the shield seated position SS. Bearing on the connector housing 222 avoids transferring force to the PCB 210, which avoids damage to a joint between the connector 220 and the PCB 210 and improves the reliability of the connector assembly 200. The tool 100 also presses the cap 240 and the shield 250 of the connector 220 successively in two stages in one cycle of the driving portion 180, efficiently terminating the wire 260 in the connector 220 while avoiding damage to the connector 220 and the PCB 210.

Claims

1. A tool for a connector assembly, comprising: a crimp head having a tool housing and a ram assembly disposed within the tool housing, the ram assembly has a connector ram and a shield ram disposed within the connector ram; and a driving portion driving movement of the ram assembly with respect to the tool housing from a retracted position to a terminated position along a longitudinal direction, the connector ram and the shield ram are movable together with respect to the tool housing from the retracted position to an intermediate position between the retracted position and the terminated position, the shield ram is movable with respect to the connector ram from the intermediate position to the terminated position of the ram assembly.
2. The tool of claim 1, wherein the ram assembly has a ram stop disposed in the tool housing, the connector ram contacts the ram stop in the intermediate position.
3. The tool of claim 2, wherein the ram stop maintains a position of the connector ram along the longitudinal direction as the shield ram moves from the intermediate position to the terminated position.
4. The tool of claim 1, wherein the driving portion has a handle and a trigger movable with respect to the handle to drive movement of the ram assembly.
5. The tool of claim 4, wherein the driving portion has a ratchet assembly restricting movement of the ram assembly back to the retracted position until the ram assembly reaches the terminated position.
6. The tool of claim 1, wherein the crimp head has a clamp assembly including a pair of jaws pivotably attached to the tool housing.
7. The tool of claim 6, wherein the ram assembly has a pair of jaw plates attached to the connector ram and movable with the connector ram, the jaw plates each abut one of the jaws and move the jaws from an open position to a closed position as the connector ram moves from the retracted

position to the intermediate position.

8. The tool of claim 7, wherein the clamp assembly has a plurality of biasing springs each having a first end disposed in the tool housing and a second end disposed in one of the jaws, the biasing springs bias the jaws toward the open position.

9. The tool of claim 1, wherein the ram assembly has an inserter rod connected to the shield ram and a shield ram adjuster disposed on an end of the inserter rod opposite the shield ram.

10. The tool of claim 9, wherein the ram assembly has a return spring disposed between the shield ram adjuster and the tool housing, the return spring biases the ram assembly toward the retracted position.

11. The tool of claim 9, wherein the ram assembly has a preload adjuster disposed around the inserter rod and a preload spring disposed between the preload adjuster and the connector ram, the driving portion overcomes a preload force of the preload spring to move the shield ram with respect to the connector ram along the longitudinal direction.

12. A terminating assembly, comprising: a connector assembly including a connector having a connector housing and a cap disposed on the connector housing; and a tool including a crimp head having a tool housing, a ram assembly disposed within the tool housing, and a clamp assembly attached to the tool housing, the ram assembly is movable with respect to the tool housing along a longitudinal direction between a retracted position and a terminated position, movement of the ram assembly from the retracted position to an intermediate position between the retracted position and the terminated position moves a pair of jaws of the clamp assembly into a closed position of the jaws in engagement with the connector housing and moves the cap to a cap seated position on the connector housing.

13. The terminating assembly of claim 12, wherein the ram assembly includes a connector ram and a shield ram disposed within the connector ram, the connector ram contacts the cap and moves the cap to the cap seated position as the ram assembly moves from the retracted position to the intermediate position.

14. The terminating assembly of claim 12, wherein the connector assembly includes a printed circuit board, the connector is disposed on the printed circuit board.

15. The terminating assembly of claim 14, wherein the connector housing has a pair of projections extending from the connector housing and the jaws each have a hook at an end opposite the tool housing, the hook of each of the jaws engages one of the pair of projections in the closed position.

16. The terminating assembly of claim 15, wherein the hook of each of the jaws is disposed between one of the pair of projections and an upper surface of the printed circuit board in the closed position.

17. The terminating assembly of claim 16, wherein the shield ram moves with respect to the connector ram from the intermediate position to the terminated position.

18. The terminating assembly of claim 17, wherein the connector ram is held by a ram stop in the tool housing at the intermediate position and remains in a same position in the tool housing along the longitudinal direction as the shield ram moves from the intermediate position to the terminated position.

19. The terminating assembly of claim 17, wherein the connector assembly includes a shield disposed on the connector housing, the shield ram contacts the shield and moves the shield to a shield seated position on the connector housing as the shield ram moves from the intermediate position to the terminated position.

20. The terminating assembly of claim 19, wherein the jaws are in the closed position in engagement with the connector housing in the intermediate position and in the terminated position.
